

Thaneesan Siventhirarajah

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Mechatronics Engineering graduate with interdisciplinary experience in product design, additive manufacturing, computer vision, robotics, and jewellery manufacturing. Skilled in CAD modeling, rapid prototyping, 3D printing and laser-based fabrication, and engineering R&D. Committed to applying research and Digital Fabrication expertise to develop innovative solutions for real-world engineering challenges.

Experience

Freelance Researcher

Self-employed

2024 - Present

Delivered projects in CAD design, robotics, computer vision, and additive manufacturing.

CAD Modeling and 3D Printing

Tensor Labs

2024 - 2026

Product prototyping, CAD/CAM and additive manufacturing.

Jewellery Manufacturing Manager

Ruby Jewellery

May 2025 - November 2026

Jewellery design, SLA 3D printing, casting, laser welding, fiber laser, and gold refining.

Research Assistant Intern

LE Robotics (Pvt) Ltd.

June 2024 - September 2024

Worked on a computer vision project focused on warehouse applications.

STEM Workshop Instructor

2022 - Present

Conducted STEM workshops for schools and personal events, focusing on robotics and engineering basics.

Education

BTech Hons (Mechatronics)

(September 2020 - September 2024)

Bachelor of Technology Honours in Science and Technology specialized in Mechatronics,
Faculty of Applied Sciences of Uva Wellassa University of Sri Lanka.

- GPA - 3.72 / 4.00 (First Class Honours)
- Medium of Instruction - English
- Effective Date of the Degree - 30 September 2024

Developing Career Skills

(2018)

Department of Technical Education & Training Sri Lanka, Ministry of Skills Development & Vocational Training

G.C.E (A/L) Physical Science Stream

(2018)

Jaffna Hindu College

Z-Score - 1.2306

Combined Mathematics - B | Physics - B | Chemistry - B

G.C.E (O/L)

(2015)

Jaffna Hindu College

Mathematics - A | Science - B | English - B | ICT - A

Research Interests

- Product Design and Development
- Robotics and Automation
- Soft Robotics and Digital Fabrication
- Additive Manufacturing, Smart Manufacturing
- Computer Vision
- Control Systems
- Advanced Kinematics
- CAD/CAM
- Jewellery Design and Manufacturing

Projects

- Identify boxes in a warehouse (Researched on various methods YOLO, U-Net, etc.)
- Voice-Controlled Prosthetic Hand with Keyword Spotting (Python, CNN, NLP, CAD Modeling and Prototype development)
- Obstacle Detection System (YOLOv8, TensorFlow, Raspberry Pi)
- Smart Chair Walking Stick (CAD Modeling and Prototype development, ESP32 and IoT based system)
- Inclined Bucket Conveyor Belt System (CAD Modeling and Prototype development, Arduino)

Publications



- Integrating Contemporary Casting Technologies in Traditional Goldsmithing: A Case Study on Gold Decker Ring Production: ISSN: 3021-6656
- YOLOv8 powered solutions for box identification in warehouses
DOI: 10.38124/ijisrt/IJISRT24SEP1017
- Voice-controlled prosthetic hand with keyword spotting: Conference: IRCUWU2024

Certifications

- Additive Manufacturing: Resin 3D Printing: LinkedIn Learning
- Rapid Prototyping for Product Design: LinkedIn Learning
- Modeling and Simulation with Simulink: Coursera
- Simulink Onramp: MathWorks
- MATLAB Onramp: MathWorks
- SOLIDWORKS CAD Design Associate (CSWA): Dassault Systèmes
- SOLIDWORKS Sheet Metal: Udemy
- SOLIDWORKS: Become a Certified Professional Today (CSWP): Udemy
- Introduction to CNC Programming: Udemy

Expertise

Computer Vision
Machine Learning
Python, C, C++

CAD Modeling
3D printing and slicing
Prototyping and Fabrication

Embedded System
Arduino, ESP32, Raspberry Pi
Jewellery manufacturing

Soft skills

Time Management
Leadership

Communication
Stress Management

Teamwork
Problem-Solving

Language

English (Professional Working)

Tamil (Native)

Sinhala (Elementary)

Honors & Awards

- Dean's list for outstanding academic performance (5 Semesters)
- LinkedEng English Camp 2021 (Participation in the Speech contest and Debate competition)
- X-mas Revamp (The most creative Category 2nd Place)

Personal Details

- Name : S. Thaneesan
- Date of Birth : 11th May 1999
- Nationality : Sri Lankan
- Gender : Male
- Marital status : Single

References

Dr. A.R.P.C.C.J. Amarasinghe
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